RCIM PAPER REFERENCE

RCIM Retuned Reference Archive Closeout Report

Retuned recovered-original model archive closeout

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Overview

This report records the closeout of recovered-original RCIM retuned model exports into the curated `models/paper_reference/rcim_retuned` archive. The closeout accepts only family-direction exports with all `20` target-level ONNX artifacts and all `20` target-level Python pickle artifacts.

Archive Completeness

Direction	Family	Retune Bundle	Eval Bundle	Export Bundle	ONNX	PKL	Exported Errors
`forward`	`SVM`	`2026-05-09-23-52-16__fw_retune_bundle`	`2026-05-09-23-52-16__fw_retune_bundle`	`2026-05-09-23-52-16__fw_retune_bundle`	`20`	`20`	`0`
`forward`	`MLP`	`2026-05-11-16-55-11__fw_retune_bundle`	`2026-05-11-16-55-11__fw_retune_bundle`	`2026-05-11-16-55-11__fw_retune_bundle`	`20`	`20`	`0`
`forward`	`RF`	`2026-05-09-10-02-19__fw_retune_bundle`	`2026-05-09-13-01-21__fw_eval_bundle`	`2026-05-09-13-01-34__fw_export_bundle`	`20`	`20`	`0`
`forward`	`DT`	`2026-05-13-15-40-16__fw_retune_bundle`	`2026-05-13-15-40-16__fw_retune_bundle`	`2026-05-13-15-40-16__fw_retune_bundle`	`20`	`20`	`0`
`forward`	`ET`	`2026-05-11-18-27-16__fw_retune_bundle`	`2026-05-11-18-27-16__fw_retune_bundle`	`2026-05-11-18-27-16__fw_retune_bundle`	`20`	`20`	`0`
`forward`	`ERT`	`2026-05-11-18-27-16__fw_retune_bundle`	`2026-05-11-18-27-16__fw_retune_bundle`	`2026-05-11-18-27-16__fw_retune_bundle`	`20`	`20`	`0`
`forward`	`GBM`	`2026-05-11-18-27-16__fw_retune_bundle`	`2026-05-11-18-27-16__fw_retune_bundle`	`2026-05-11-18-27-16__fw_retune_bundle`	`20`	`20`	`0`
`forward`	`HGBM`	`2026-05-11-18-27-16__fw_retune_bundle`	`2026-05-12-18-16-50__fw_eval_bundle`	`2026-05-12-18-20-41__fw_export_bundle`	`20`	`20`	`0`
`forward`	`XGBM`	`2026-05-11-16-55-11__fw_retune_bundle`	`2026-05-11-16-55-11__fw_retune_bundle`	`2026-05-11-16-55-11__fw_retune_bundle`	`20`	`20`	`0`
`forward`	`LGBM`	`2026-05-12-11-20-54__fw_retune_bundle`	`2026-05-12-11-20-54__fw_retune_bundle`	`2026-05-12-11-20-54__fw_retune_bundle`	`20`	`20`	`0`
`forward`	`ELM`	`2026-05-11-18-21-23__fw_retune_bundle`	`2026-05-11-18-21-23__fw_retune_bundle`	`2026-05-11-18-21-23__fw_retune_bundle`	`20`	`20`	`0`
`backward`	`SVM`	`2026-05-09-23-42-54__bw_retune_bundle`	`2026-05-09-23-42-54__bw_retune_bundle`	`2026-05-09-23-42-54__bw_retune_bundle`	`20`	`20`	`0`
`backward`	`MLP`	`2026-05-11-09-19-07__bw_retune_bundle`	`2026-05-11-09-19-07__bw_retune_bundle`	`2026-05-11-09-19-07__bw_retune_bundle`	`20`	`20`	`0`
`backward`	`RF`	`2026-05-09-09-23-24__bw_retune_bundle`	`2026-05-09-12-31-49__bw_eval_bundle`	`2026-05-09-12-32-07__bw_export_bundle`	`20`	`20`	`0`
`backward`	`DT`	`2026-05-09-09-21-57__bw_retune_bundle`	`2026-05-09-09-21-57__bw_retune_bundle`	`2026-05-09-09-21-57__bw_retune_bundle`	`20`	`20`	`0`
`backward`	`ET`	`2026-05-12-11-08-07__bw_retune_bundle`	`2026-05-12-11-08-07__bw_retune_bundle`	`2026-05-12-11-08-07__bw_retune_bundle`	`20`	`20`	`0`
`backward`	`ERT`	`2026-05-12-11-08-07__bw_retune_bundle`	`2026-05-12-11-08-07__bw_retune_bundle`	`2026-05-12-11-08-07__bw_retune_bundle`	`20`	`20`	`0`
`backward`	`GBM`	`2026-05-12-11-08-07__bw_retune_bundle`	`2026-05-12-11-08-07__bw_retune_bundle`	`2026-05-12-11-08-07__bw_retune_bundle`	`20`	`20`	`0`
`backward`	`HGBM`	`2026-05-12-11-08-07__bw_retune_bundle`	`2026-05-12-18-22-07__bw_eval_bundle`	`2026-05-12-18-22-27__bw_export_bundle`	`20`	`20`	`0`
`backward`	`XGBM`	`2026-05-11-09-19-07__bw_retune_bundle`	`2026-05-11-09-19-07__bw_retune_bundle`	`2026-05-11-09-19-07__bw_retune_bundle`	`20`	`20`	`0`
`backward`	`LGBM`	`2026-05-11-16-05-54__bw_retune_bundle`	`2026-05-11-16-05-54__bw_retune_bundle`	`2026-05-11-16-05-54__bw_retune_bundle`	`20`	`20`	`0`

Direction	Family	Retune Bundle	Eval Bundle	Export Bundle	ONNX	PKL	Exported Errors
backward	ELM	2026-05-11-09-19-07__bw_retune_bundle	2026-05-11-11-38-37__bw_eval_bundle	2026-05-11-11-38-49__bw_export_bundle	20	20	0

Mean Evaluation Metrics

These values come from the accepted `Eval` stage `summaryCrossValidation+_3.8_allFreq.csv` files.

Direction	Family	MSE	RMSE	MAE	MAPE
forward	SVM	0.45221	0.2978	0.184519	0.763904
forward	MLP	0.383808	0.29569	0.213553	66411031743924.046875
forward	RF	0.112274	0.145542	0.055486	0.177337
forward	DT	0.155008	0.170755	0.063618	0.274174
forward	ET	0.140874	0.172043	0.067927	0.227937
forward	ERT	0.096401	0.138521	0.055956	0.191636
forward	GBM	0.123025	0.15127	0.066212	0.239887
forward	HGBM	0.125222	0.152528	0.077089	0.42525
forward	XGBM	0.102465	0.139981	0.060535	0.318981
forward	LGBM	0.127166	0.153622	0.077013	0.348775
forward	ELM	0.430192	0.289775	0.211945	1.171391
backward	SVM	0.469998	0.340568	0.212199	0.685275
backward	MLP	0.31688	0.31866	0.207254	147176760816685.3125
backward	RF	0.067836	0.123949	0.047092	0.245926
backward	DT	0.11403	0.164084	0.060589	0.370306
backward	ET	0.144024	0.18662	0.061084	0.274356
backward	ERT	0.056633	0.118571	0.047711	0.241712
backward	GBM	0.202892	0.222167	0.080926	0.35952
backward	HGBM	0.110792	0.16732	0.085725	0.380283
backward	XGBM	0.260145	0.255916	0.172579	0.763643
backward	LGBM	0.186807	0.218703	0.145023	0.675751
backward	ELM	0.387312	0.313587	0.215718	0.799429

Forward Retuned Tables

Table 2 - Amplitude MAE

Model	0	1	3	39	40	78	81	156	162	240
SVM	0.002912	4.66302e-05	6.04508e-05	7.55757e-05	7.10815e-05	0.000121731	5.12823e-05	0.000696237	0.000698217	0.00024679
MLP	0.014562	0.007778	0.0111	0.007793	0.009778	0.011258	0.011897	0.008305	0.00701	0.015629
RF	0.00312	2.69224e-05	1.92205e-05	2.61055e-05	2.2637e-05	3.59297e-05	1.05759e-05	4.57812e-05	5.27089e-05	3.28826e-05
DT	0.00351	3.10238e-05	2.31505e-05	3.6618e-05	2.97753e-05	5.5576e-05	1.52993e-05	8.91254e-05	6.69908e-05	4.50935e-05
ET	0.003923	3.39405e-05	2.73594e-05	3.96039e-05	2.81046e-05	6.39258e-05	1.5678e-05	6.23185e-05	6.8023e-05	4.45344e-05
ERT	0.003262	2.51681e-05	2.08696e-05	2.6821e-05	2.21245e-05	3.62432e-05	1.09854e-05	4.05968e-05	5.01524e-05	3.70563e-05
GBM	0.002887	2.68313e-05	2.06267e-05	2.64032e-05	2.42143e-05	3.83771e-05	1.19535e-05	0.000151854	0.000203692	5.85112e-05
HGBM	0.002618	2.63076e-05	1.86476e-05	2.34801e-05	2.56511e-05	2.36904e-05	1.22957e-05	0.000107198	0.000119594	3.49172e-05
XGBM	0.002842	5.29274e-05	5.17414e-05	6.83342e-05	6.44126e-05	8.84949e-05	4.65581e-05	0.000135924	0.000104757	0.000115468
LGBM	0.002575	2.62664e-05	1.87327e-05	2.33772e-05	2.5798e-05	2.51835e-05	1.20466e-05	0.000107466	0.000118095	3.52105e-05
ELM	0.008394	2.88142e-05	7.87396e-05	0.000112282	4.01642e-05	0.000296962	2.83329e-05	0.000856622	0.001067	0.000369496

Table 3 - Amplitude RMSE

Model	0	1	3	39	40	78	81	156	162	240
SVM	0.004123	6.08873e-05	7.62318e-05	9.55895e-05	8.97431e-05	0.000165775	6.03122e-05	0.001528	0.002377	0.000612235
MLP	0.023068	0.013469	0.02176	0.015826	0.018224	0.021632	0.020561	0.015097	0.014136	0.025682
RF	0.004098	3.81916e-05	2.78872e-05	3.62718e-05	3.33471e-05	5.61394e-05	1.8065e-05	0.000163705	0.000154321	5.64108e-05
DT	0.004879	4.30523e-05	3.33725e-05	5.05612e-05	4.50974e-05	7.97431e-05	2.38369e-05	0.000298278	0.000196766	7.18479e-05
ET	0.005512	4.98162e-05	4.46899e-05	5.37782e-05	4.20959e-05	8.92871e-05	2.55713e-05	0.000190094	0.000186599	6.97557e-05
ERT	0.004164	3.5925e-05	3.07983e-05	3.76465e-05	3.28386e-05	5.4169e-05	1.87404e-05	0.000144831	0.000158585	6.67768e-05
GBM	0.004091	3.87048e-05	2.7237e-05	3.78415e-05	3.57058e-05	5.40153e-05	2.03112e-05	0.000471228	0.000738856	0.00015009
HGBM	0.003898	3.66942e-05	2.64492e-05	3.2727e-05	3.5704e-05	3.51107e-05	1.92161e-05	0.000264253	0.000275736	7.01425e-05
XGBM	0.003956	6.81352e-05	6.50398e-05	8.87521e-05	8.33804e-05	0.000116671	5.98269e-05	0.000336537	0.000236608	0.000164157
LGBM	0.00379	3.66354e-05	2.67199e-05	3.21329e-05	3.57956e-05	3.707e-05	1.89939e-05	0.000267423	0.000266197	7.02233e-05
ELM	0.010659	3.82909e-05	9.6453e-05	0.000153107	6.26075e-05	0.0003916	3.88056e-05	0.00157	0.002357	0.000636913

Table 4 - Phase MAE

Model	1	3	39	40	78	81	156	162	240
SVM	0.002861	0.039126	0.038193	0.095298	0.19446	0.167816	1.545304	0.854949	0.747385
MLP	0.020549	0.068079	0.081528	0.07124	0.145159	0.141555	1.653017	0.90848	0.781412
RF	0.001953	0.024688	0.02624	0.036396	0.054711	0.045866	0.410856	0.234445	0.271166
DT	0.002239	0.02734	0.032261	0.045456	0.07449	0.06362	0.490408	0.245312	0.287325
ET	0.002979	0.032667	0.032921	0.046145	0.105701	0.077808	0.474651	0.284212	0.297142
ERT	0.002328	0.028784	0.027723	0.03521	0.062335	0.051688	0.400729	0.233554	0.273227
GBM	0.002203	0.023329	0.025456	0.037337	0.050657	0.047991	0.479103	0.298957	0.355748
HGBM	0.001955	0.025438	0.020247	0.037234	0.071251	0.048207	0.605127	0.337684	0.391634
XGBM	0.0019	0.025124	0.020946	0.035046	0.068123	0.043829	0.470111	0.251684	0.290375
LGBM	0.001928	0.025764	0.020305	0.03646	0.073436	0.045802	0.603746	0.346318	0.383523
ELM	0.002771	0.080557	0.089368	0.067536	0.19027	0.174713	1.703741	1.116297	0.80237

Table 5 - Phase RMSE

Model	1	3	39	40	78	81	156	162	240
SVM	0.00436	0.054297	0.060574	0.132484	0.362591	0.241217	2.02453	1.654005	1.412756
MLP	0.032351	0.086326	0.104059	0.103366	0.212569	0.179317	1.994397	1.406654	1.202815
RF	0.002714	0.034748	0.046435	0.054608	0.135891	0.06584	0.967807	0.762623	0.835498
DT	0.003115	0.039113	0.060054	0.070673	0.151314	0.095694	1.225799	0.891606	0.872011
ET	0.004395	0.049717	0.05807	0.071786	0.294988	0.130349	1.092873	0.927118	0.805299
ERT	0.003522	0.040693	0.053397	0.056126	0.156387	0.082454	0.876494	0.726699	0.769894
GBM	0.003227	0.033096	0.041325	0.056051	0.112009	0.075405	0.953229	0.825631	0.919771
HGBM	0.002681	0.036307	0.032742	0.057628	0.139074	0.070305	1.03431	0.784113	0.888707
XGBM	0.002489	0.03353	0.034166	0.056416	0.147066	0.066047	0.896522	0.773625	0.784581
LGBM	0.002673	0.03642	0.032488	0.056709	0.145076	0.067668	1.049463	0.794237	0.883119
ELM	0.004207	0.099629	0.112824	0.098927	0.285837	0.233381	2.031703	1.636152	1.276826

Backward Retuned Tables

Table 2 - Amplitude MAE

Model	0	1	3	39	40	78	81	156	162	240
SVM	0.003417	7.05621e-05	6.44134e-05	0.000108184	3.99598e-05	0.000140876	3.55141e-05	0.000759711	0.000512494	0.000342361
MLP	0.016062	0.008565	0.02068	0.009938	0.009976	0.007135	0.010629	0.008277	0.011534	0.007464
RF	0.002937	2.37386e-05	2.07275e-05	1.72732e-05	2.76117e-05	5.1867e-05	1.00356e-05	0.000117828	9.1534e-05	6.23487e-05
DT	0.003504	2.69959e-05	3.42058e-05	2.59035e-05	3.23554e-05	0.000105717	1.36692e-05	0.000191057	0.000165638	0.000125839
ET	0.003322	2.65069e-05	2.63791e-05	2.03945e-05	3.03482e-05	7.00267e-05	9.3832e-06	0.000111551	9.2988e-05	0.000129119
ERT	0.003049	2.17166e-05	2.08911e-05	1.70013e-05	2.7271e-05	4.75824e-05	9.04573e-06	0.000102756	9.06934e-05	8.63274e-05
GBM	0.003907	2.9644e-05	2.4405e-05	2.18322e-05	3.7928e-05	8.77184e-05	1.31074e-05	0.000407784	0.00013776	0.00020224
HGBM	0.003336	3.32158e-05	2.92694e-05	2.23226e-05	2.9316e-05	8.58519e-05	1.20384e-05	0.000355944	0.000276624	0.000115003
XGBM	0.009006	5.01457e-05	6.59465e-05	6.57939e-05	4.6482e-05	0.000306515	2.74418e-05	0.000887201	0.000532549	0.000212008
LGBM	0.006586	3.53015e-05	4.62261e-05	4.03759e-05	3.28733e-05	0.000230488	1.56541e-05	0.000577349	0.000432744	0.000209221
ELM	0.006105	4.0656e-05	6.19064e-05	6.94347e-05	3.74189e-05	0.000314255	2.18478e-05	0.001246	0.000861271	0.000374497

Table 3 - Amplitude RMSE

Model	0	1	3	39	40	78	81	156	162	240
SVM	0.004321	8.92984e-05	7.59251e-05	0.000122372	5.24986e-05	0.00018634	4.13104e-05	0.002147	0.001733	0.000768566
MLP	0.023489	0.014012	0.034222	0.016157	0.019279	0.01215	0.018511	0.011909	0.020039	0.01354
RF	0.003822	4.01928e-05	3.95792e-05	2.41367e-05	4.08393e-05	7.35254e-05	1.45444e-05	0.000351421	0.000350566	0.000143677
DT	0.004755	3.89166e-05	4.63022e-05	3.5792e-05	4.65966e-05	0.000141259	2.1564e-05	0.000458745	0.000633652	0.000247847
ET	0.004572	3.78036e-05	4.05121e-05	3.04868e-05	4.2643e-05	9.56236e-05	1.38535e-05	0.000333265	0.000508611	0.000340328
ERT	0.004034	3.32592e-05	3.37697e-05	2.42355e-05	3.99313e-05	6.91306e-05	1.34407e-05	0.000318311	0.000352948	0.000173187
GBM	0.005304	4.27122e-05	3.50066e-05	3.02839e-05	5.45847e-05	0.000114522	2.03896e-05	0.002273	0.00066182	0.000854605
HGBM	0.004305	5.05026e-05	4.01874e-05	3.17414e-05	4.19043e-05	0.000115062	1.73374e-05	0.00091268	0.000819282	0.000242704
XGBM	0.011001	7.54328e-05	8.55087e-05	8.04608e-05	5.83e-05	0.00036438	3.40776e-05	0.001727	0.001146	0.000500722
LGBM	0.008224	5.37816e-05	5.75288e-05	5.02789e-05	4.44481e-05	0.000274288	2.08906e-05	0.00103	0.000972133	0.000415477
ELM	0.007959	6.49451e-05	8.87046e-05	8.60459e-05	4.79292e-05	0.000405392	2.81136e-05	0.002354	0.001667	0.000692278

Table 4 - Phase MAE

Model	1	3	39	40	78	81	156	162	240
SVM	0.003277	0.030147	1.533948	0.239405	0.16531	0.165708	0.519181	0.647235	0.934282
MLP	0.012577	0.047895	0.883528	0.223007	0.117947	0.164507	0.550536	0.633432	0.74779
RF	0.004475	0.020274	0.259613	0.088339	0.055395	0.100364	0.103195	0.104946	0.201875
DT	0.002297	0.030949	0.264166	0.109481	0.104082	0.131394	0.154214	0.138677	0.272296
ET	0.002216	0.024758	0.251627	0.12914	0.061557	0.117384	0.137711	0.158073	0.335381
ERT	0.001928	0.022496	0.224182	0.093953	0.05315	0.092693	0.138368	0.099516	0.224467
GBM	0.001997	0.033333	0.32552	0.111136	0.060508	0.119325	0.33279	0.205319	0.423721
HGBM	0.004355	0.031669	0.48239	0.11461	0.089437	0.116216	0.251079	0.198438	0.422011
XGBM	0.007236	0.105	1.368352	0.237439	0.149121	0.154242	0.394363	0.2798	0.74483
LGBM	0.003988	0.075939	1.007733	0.159638	0.119033	0.16781	0.347521	0.403988	0.606608
ELM	0.005351	0.095946	1.401709	0.21804	0.14142	0.191291	0.566196	0.85855	0.826733

Table 5 - Phase RMSE

Model	1	3	39	40	78	81	156	162	240
SVM	0.004976	0.050379	1.904003	0.390645	0.302424	0.208329	0.96709	1.234879	1.739106
MLP	0.019934	0.061518	1.488289	0.387442	0.157908	0.216736	0.922763	1.028735	1.240815
RF	0.017312	0.031548	0.925908	0.133734	0.089557	0.134517	0.321643	0.324612	0.495247
DT	0.00395	0.046179	1.158681	0.159423	0.153046	0.187845	0.411266	0.488621	0.666233
ET	0.00377	0.037153	1.108501	0.245136	0.110455	0.16568	0.50731	0.556874	0.991502
ERT	0.002972	0.034339	0.766994	0.159162	0.095696	0.127265	0.383862	0.279082	0.516958
GBM	0.00295	0.044645	1.141662	0.180293	0.107463	0.16886	0.988349	0.695553	1.10418
HGBM	0.008257	0.042245	1.003621	0.175671	0.164139	0.159327	0.552074	0.448768	0.785715
XGBM	0.011072	0.123145	1.59109	0.382244	0.244668	0.191574	0.767348	0.565452	1.226653
LGBM	0.006405	0.089842	1.316174	0.239411	0.186768	0.208583	0.68118	0.644011	0.990548
ELM	0.011728	0.12662	1.852693	0.376851	0.186686	0.249883	0.946219	1.157153	1.350511

Source Recovery Notes

- `RF` uses separate backward `Eval` and `Export` recovery bundles after the parser fix.
- `ELM` uses later recovery export bundles where the custom ONNX exporter is active.
- `GBM` uses the original multi-family retune bundle because that bundle contains the accepted `GBM` eval row and complete exports.
- `HGBM` uses later recovery export bundles that removed earlier ONNX export errors.
- `LGBM` uses the quieter retune factory surface so native LightGBM chatter does not bury failure evidence.